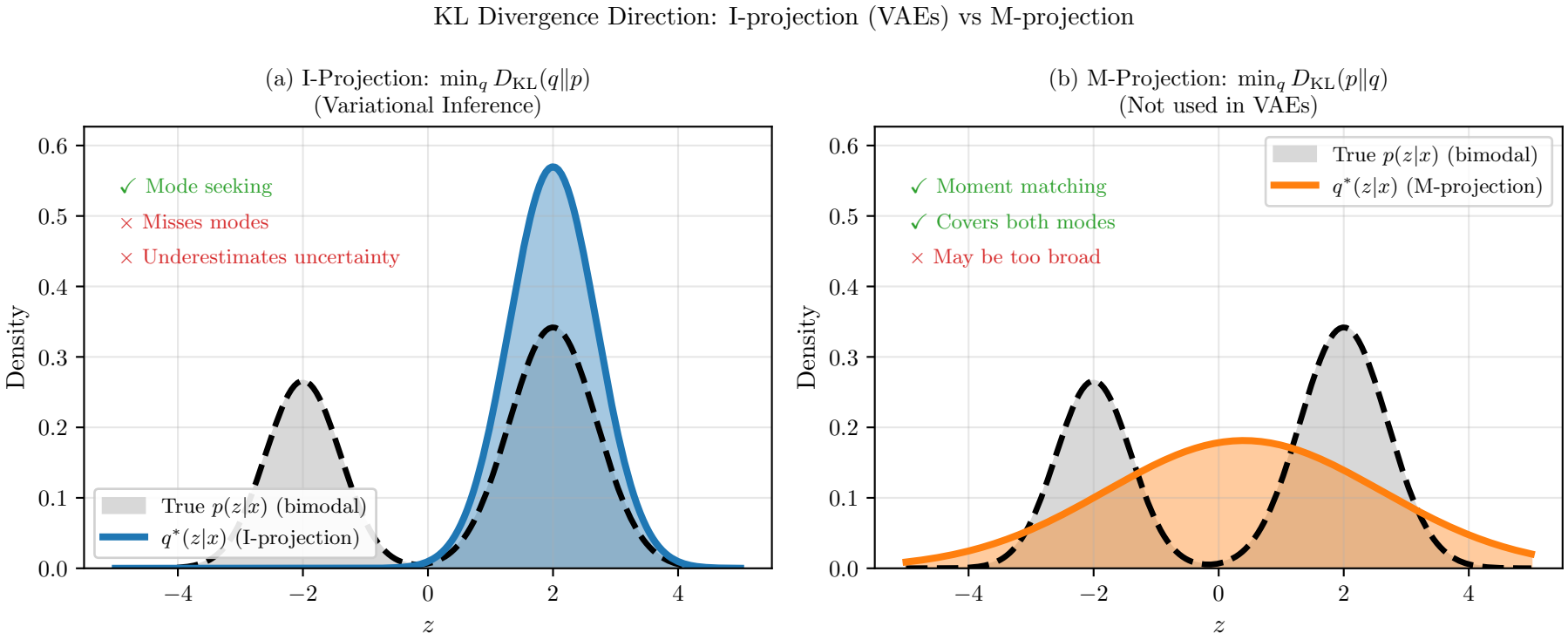
KL Divergence Direction: I-projection (VAEs) vs M-projection
(a) I-Projection: $\min_q D_{\mathrm{KL}}(q\|p)$
(Variational Inference)
✓ Mode seeking
× Misses modes
× Underestimates uncertainty
True $p(z|x)$ (bimodal)
$q^*(z|x)$ (I-projection)
Density
z
(b) M-Projection: $\min_q D_{\mathrm{KL}}(p\|q)$
(Not used in VAEs)
✓ Moment matching
✓ Covers both modes
× May be too broad
True $p(z|x)$ (bimodal)
$q^*(z|x)$ (M-projection)
Density
z